Mathematics of Machine Learning

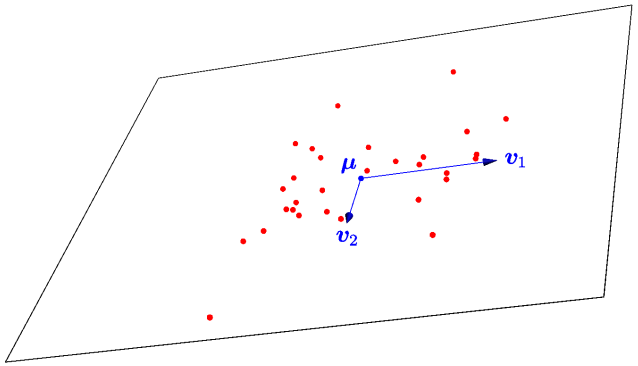
Matrix Factorisation

Name	Titanic	Dr No	The Matrix	Spartacus	Ten
Alice	2	∅	4	5	2
Bob	5	3	∅	3	∅
Chas	∅	2	3	5	3

Matrix completion, low-rank approximation, recommender systems, alternating least squares, non-negative matrix factorisation

Outline

1. **Missing Data and Recommenders**
2. Low-Rank Approximation
3. Probabilistic Approach
4. Non-Negative Matrix Factorisation



Matrix Completion

- It is extremely common to be given a matrix of data with missing entries (particularly when the data comes from humans filling in forms)
- Most learning machines need a complete set of data, so we either throw away examples or features with missing values (sometimes leaving nothing) or do our best to infer them
- Inferring them only works if there is structure in the matrix that we can exploit
- The structure we use in this lecture: **structured data typically lies close to a low-dimensional subspace** (the idea behind PCA and the SVD)

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Recommender Systems

- The best-known example: recommending items to users, which can be financially important
  - ⋆ Recommendations are reported to drive a large share of Amazon's sales
  - ⋆ Netflix (then a DVD rental company, now a streaming service) ran a $1 million competition (2006–09) to improve the quality of its recommendations
- **Content-based filtering** uses (often text) features to find similar items; **collaborative filtering** uses ratings supplied (implicitly or explicitly) by users
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Recommending Movies

- The classic example is to recommend a movie given some ratings from users

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- Ø means there are no ratings
- The task is to predict the unknown ratings and make a recommendation
- Typically data sets are huge and very sparse

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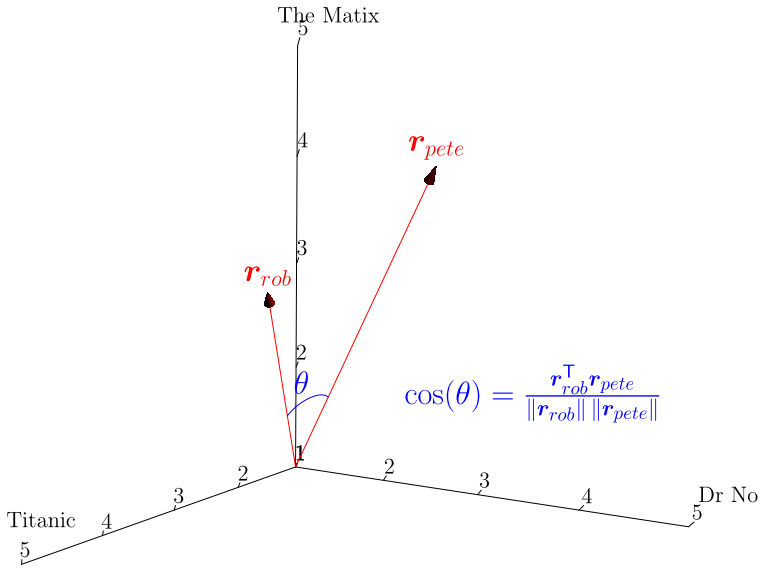
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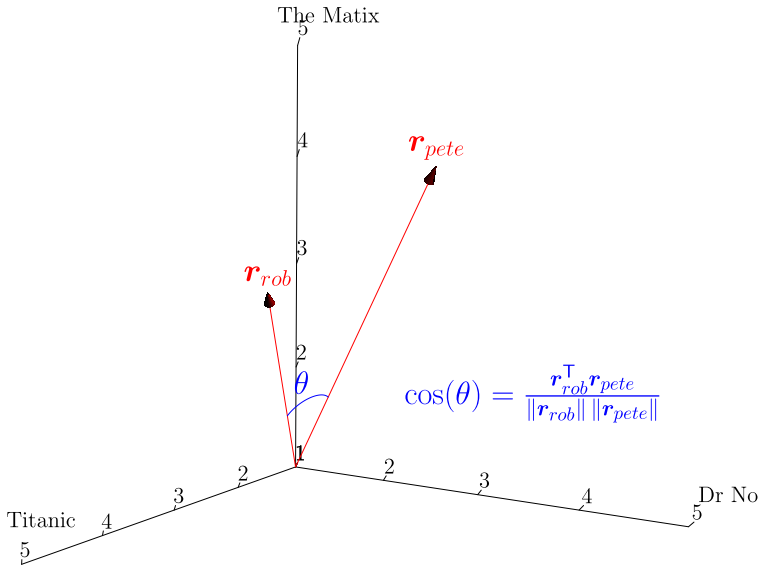
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- Use the recommendation of similar users weighted by their similarity
- Can also recommend items that are similar (i.e. items that have received similar ratings) to those that a user rates highly



$$\hat{R}_{ui} = \frac{\sum_{u'} \text{sim}(u, u') R_{u'i}}{\sum_{u'} \text{sim}(u, u')}$$

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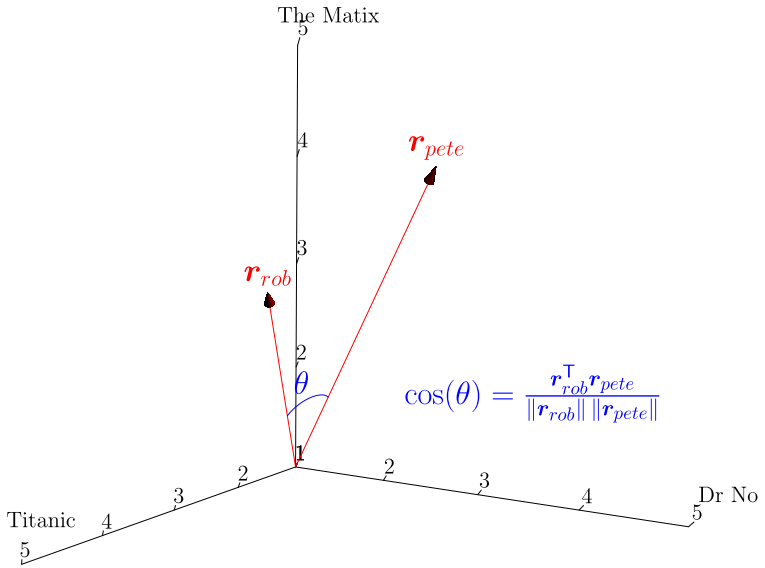
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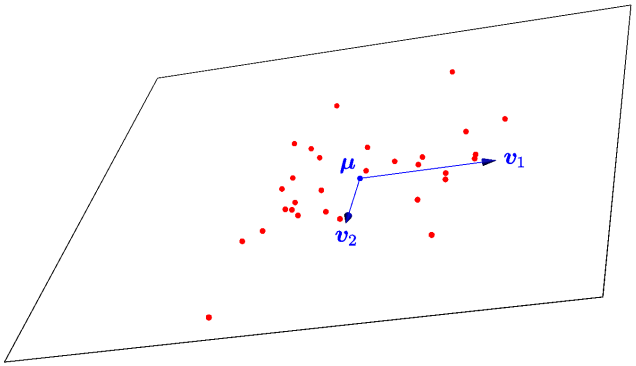
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Replace by Means

- The simplest guess uses the means, column-wise and row-wise

$$R_{ui} \approx \bar{R}_u + \bar{R}_i - \bar{R}$$

  where

  - $\bar{R}_u$ is the mean rating of user u
  - $\bar{R}_i$ is the mean rating for item i
  - $\bar{R}$ is the mean overall rating

- Gets the means right and gives a reasonable rating
- (For general missing data: replace by the feature's mean)
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Best Low-Rank Approximation

- Recall the SVD of a complete matrix, with $s_1 \geq s_2 \geq \cdots \geq 0$

$$\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\mathsf{T} = \sum_k s_k \boldsymbol{u}_k \boldsymbol{v}_k^\mathsf{T}$$

- Keeping the K largest singular values gives a rank-K matrix $\mathbf{X}_K = \sum_{k=1}^{K} s_k \boldsymbol{u}_k \boldsymbol{v}_k^\mathsf{T}$
- **Eckart–Young theorem**: $\mathbf{X}_K$ is the best rank-K approximation to $\mathbf{X}$, with error $\|\mathbf{X} - \mathbf{X}_K\|_F^2 = \sum_{k>K} s_k^2$ (also best in the spectral norm, with error s_{K+1})
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Completing a Matrix with the SVD

- The SVD needs a complete matrix, so we iterate

  1. Fill in the missing entries (e.g. with the user means)
  2. Compute the best rank-K approximation $\mathbf{L}$ of the filled-in matrix
  3. Replace the missing entries (only) by the values in $\mathbf{L}$
  4. Repeat from 2

- With a mask $\boldsymbol{W}$ ($W_{ij} = 1$ where X_{ij} is observed, 0 otherwise) and $\odot$ the element-by-element product

$$\mathbf{X}(t+1) = \boldsymbol{W} \odot \mathbf{X} + \overline{\boldsymbol{W}} \odot \mathbf{L}(t)$$

  where $\overline{\boldsymbol{W}} = 1 - \boldsymbol{W}$

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SVD in Operation

$\mathbf{X}$: observed ratings, ? = missing

$$\begin{pmatrix} 2.1 & 2.5 & 2.2 & 3.9 & ? & 4.2 & ? & 2.3 \\ 3.2 & 3.0 & 2.8 & 3.3 & ? & 3.6 & ? & 2.9 \\ ? & 2.1 & 2.0 & 4.4 & 2.1 & 4.7 & 4.3 & 1.9 \\ 3.5 & 3.6 & 3.9 & 2.9 & 3.4 & 2.8 & 2.7 & 3.2 \\ 4.1 & ? & 4.2 & 2.1 & 4.2 & ? & ? & 3.9 \\ 4.4 & ? & 4.6 & 2.2 & 4.7 & 2.2 & 1.7 & ? \end{pmatrix}$$

SVD in Operation

$$\mathbf{X}(0) = \boldsymbol{W} \odot \mathbf{X} + \overline{\boldsymbol{W}} \odot (\boldsymbol{\mu}\, \mathbf{1}^{\mathsf{T}}) \text{ (user means)}$$

$$\begin{pmatrix}
2.1 & 2.5 & 2.2 & 3.9 & 2.9 & 4.2 & 2.9 & 2.3 \\
3.2 & 3.0 & 2.8 & 3.3 & 3.1 & 3.6 & 3.1 & 2.9 \\
3.1 & 2.1 & 2.0 & 4.4 & 2.1 & 4.7 & 4.3 & 1.9 \\
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4.4 & 3.3 & 4.6 & 2.2 & 4.7 & 2.2 & 1.7 & 3.3
\end{pmatrix}$$

RMS error on hidden entries: 0.95

SVD in Operation

$\mathbf{L}(0)$: best rank-2 approximation of $\mathbf{X}(0)$

$$\begin{pmatrix} 2.6 & 2.4 & 2.2 & 3.6 & 2.5 & 4.0 & 3.4 & 2.2 \\ 3.2 & 2.9 & 3.0 & 3.3 & 3.1 & 3.7 & 3.2 & 2.7 \\ 2.6 & 2.4 & 1.9 & 4.4 & 2.3 & 4.8 & 4.1 & 2.1 \\ 3.6 & 3.2 & 3.7 & 2.6 & 3.7 & 3.1 & 2.7 & 3.2 \\ 4.2 & 3.7 & 4.3 & 2.9 & 4.3 & 3.4 & 3.1 & 3.6 \\ 4.1 & 3.6 & 4.6 & 1.8 & 4.4 & 2.2 & 2.1 & 3.6 \end{pmatrix}$$

SVD in Operation

$$\mathbf{X}(1) = \boldsymbol{W} \odot \mathbf{X} + \overline{\boldsymbol{W}} \odot \mathbf{L}(0)$$

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RMS error on hidden entries: 0.61

SVD in Operation

$\mathbf{L}(1)$: best rank-2 approximation of $\mathbf{X}(1)$

$$\begin{pmatrix} 2.4 & 2.4 & 2.2 & 3.8 & 2.3 & 4.1 & 3.6 & 2.2 \\ 3.1 & 2.9 & 3.0 & 3.3 & 3.1 & 3.6 & 3.2 & 2.8 \\ 2.3 & 2.3 & 1.9 & 4.4 & 2.1 & 4.7 & 4.2 & 2.1 \\ 3.6 & 3.3 & 3.7 & 2.7 & 3.7 & 3.1 & 2.7 & 3.2 \\ 4.1 & 3.8 & 4.3 & 2.7 & 4.3 & 3.1 & 2.7 & 3.7 \\ 4.3 & 3.8 & 4.6 & 1.9 & 4.5 & 2.3 & 1.9 & 3.8 \end{pmatrix}$$

SVD in Operation

$$\mathbf{X}(2) = \boldsymbol{W} \odot \mathbf{X} + \overline{\boldsymbol{W}} \odot \mathbf{L}(1)$$

$$\begin{pmatrix} 2.1 & 2.5 & 2.2 & 3.9 & 2.3 & 4.2 & 3.6 & 2.3 \\ 3.2 & 3.0 & 2.8 & 3.3 & 3.1 & 3.6 & 3.2 & 2.9 \\ 2.3 & 2.1 & 2.0 & 4.4 & 2.1 & 4.7 & 4.3 & 1.9 \\ 3.5 & 3.6 & 3.9 & 2.9 & 3.4 & 2.8 & 2.7 & 3.2 \\ 4.1 & 3.8 & 4.2 & 2.1 & 4.2 & 3.1 & 2.7 & 3.9 \\ 4.4 & 3.8 & 4.6 & 2.2 & 4.7 & 2.2 & 1.7 & 3.8 \end{pmatrix}$$

RMS error on hidden entries: 0.41

SVD in Operation

$\mathbf{L}(2)$: best rank-2 approximation of $\mathbf{X}(2)$

$$\begin{pmatrix} 2.3 & 2.4 & 2.2 & 3.9 & 2.3 & 4.2 & 3.7 & 2.2 \\ 3.1 & 3.0 & 3.0 & 3.3 & 3.1 & 3.6 & 3.2 & 2.8 \\ 2.2 & 2.3 & 1.9 & 4.4 & 2.1 & 4.7 & 4.3 & 2.0 \\ 3.6 & 3.4 & 3.7 & 2.7 & 3.7 & 3.0 & 2.7 & 3.3 \\ 4.1 & 3.8 & 4.3 & 2.6 & 4.2 & 2.9 & 2.5 & 3.7 \\ 4.4 & 4.0 & 4.6 & 1.9 & 4.5 & 2.2 & 1.9 & 3.9 \end{pmatrix}$$

SVD in Operation

$$\mathbf{X}(3) = \boldsymbol{W} \odot \mathbf{X} + \overline{\boldsymbol{W}} \odot \mathbf{L}(2)$$

$$\begin{pmatrix} 2.1 & 2.5 & 2.2 & 3.9 & 2.3 & 4.2 & 3.7 & 2.3 \\ 3.2 & 3.0 & 2.8 & 3.3 & 3.1 & 3.6 & 3.2 & 2.9 \\ 2.2 & 2.1 & 2.0 & 4.4 & 2.1 & 4.7 & 4.3 & 1.9 \\ 3.5 & 3.6 & 3.9 & 2.9 & 3.4 & 2.8 & 2.7 & 3.2 \\ 4.1 & 3.8 & 4.2 & 2.1 & 4.2 & 2.9 & 2.5 & 3.9 \\ 4.4 & 4.0 & 4.6 & 2.2 & 4.7 & 2.2 & 1.7 & 3.9 \end{pmatrix}$$

RMS error on hidden entries: 0.29

SVD in Operation

$\mathbf{L}(3)$: best rank-2 approximation of $\mathbf{X}(3)$

$$\begin{pmatrix} 2.3 & 2.4 & 2.2 & 3.9 & 2.3 & 4.2 & 3.8 & 2.2 \\ 3.0 & 3.0 & 3.0 & 3.3 & 3.1 & 3.6 & 3.2 & 2.8 \\ 2.1 & 2.2 & 1.9 & 4.4 & 2.0 & 4.7 & 4.3 & 2.0 \\ 3.6 & 3.4 & 3.7 & 2.7 & 3.6 & 3.0 & 2.6 & 3.3 \\ 4.1 & 3.9 & 4.3 & 2.5 & 4.2 & 2.7 & 2.3 & 3.8 \\ 4.4 & 4.1 & 4.6 & 2.0 & 4.5 & 2.2 & 1.9 & 4.0 \end{pmatrix}$$

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$$\mathbf{X}(4) = \boldsymbol{W} \odot \mathbf{X} + \overline{\boldsymbol{W}} \odot \mathbf{L}(3)$$

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RMS error on hidden entries: 0.22

SVD in Operation

True values of the hidden entries

$$\begin{pmatrix} 2.1 & 2.5 & 2.2 & 3.9 & 2.3 & 4.2 & 4.2 & 2.3 \\ 3.2 & 3.0 & 2.8 & 3.3 & 3.1 & 3.6 & 3.4 & 2.9 \\ 1.9 & 2.1 & 2.0 & 4.4 & 2.1 & 4.7 & 4.3 & 1.9 \\ 3.5 & 3.6 & 3.9 & 2.9 & 3.4 & 2.8 & 2.7 & 3.2 \\ 4.1 & 4.1 & 4.2 & 2.1 & 4.2 & 2.6 & 2.3 & 3.9 \\ 4.4 & 4.4 & 4.6 & 2.2 & 4.7 & 2.2 & 1.7 & 4.3 \end{pmatrix}$$

final RMS error 0.22 (from 0.95 with means)

Low-Rank Approximation

- More directly, approximate an $m \times n$ matrix $\mathbf{X}$ by

$$\mathbf{X} \approx \mathbf{A}\,\mathbf{B}$$

  where $\mathbf{A}$ is $m \times K$ and $\mathbf{B}$ is $K \times n$

- $\mathbf{AB}$ has $rank$ at most K
- We replace $m \times n$ elements by $(m+n) \times K$ free parameters
- K controls the complexity of the model: large K fits the data better but may over-fit
- Choose K by cross-validation (hide some of the observed entries and see how well we predict them)

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Low-Rank Updates: LoRA

- **LoRA** (low-rank adaptation, Hu et al. 2021) uses the same idea to fine-tune a large network: a pretrained weight matrix $\boldsymbol{W}_0$ $(m \times n)$ becomes $\boldsymbol{W}_0 + \Delta\boldsymbol{W}$

- LoRA freezes $\boldsymbol{W}_0$ and learns only a low-rank change

$$\Delta\boldsymbol{W} = \mathbf{A}\mathbf{B}, \qquad \mathbf{A}\ (m \times K),\ \mathbf{B}\ (K \times n),\ K \ll m,n$$

  with one factor starting at zero (so training starts from $\boldsymbol{W}_0$)

- For $m = n = 4096$ and $K = 8$: $(m+n)K \approx 6.6 \times 10^4$ parameters to train instead of $mn \approx 1.7 \times 10^7$ (0.4%)

- The bet: the change a new task needs is close to low rank; each task stores only its own $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{AB}$ can be added into $\boldsymbol{W}_0$ afterwards

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Least-Squares

- Assume we have elements of $\mathbf{X}$ only at the locations $\mathcal{D}$
- We can seek a least-squares solution for $\mathbf{A}$ and $\mathbf{B}$

$$(\mathbf{A},\mathbf{B}) = \underset{(\mathbf{A},\mathbf{B})}{\operatorname{argmin}} \sum_{(i,j)\in\mathcal{D}} \left(X_{ij} - \sum_{k=1}^{K} A_{ik} B_{kj} \right)^2$$

- OK, but it can over-fit
- $\mathbf{A}$ and $\mathbf{B}$ are never unique: $\mathbf{AB} = (\mathbf{AG})(\mathbf{G}^{-1}\mathbf{B})$ for any invertible $K \times K$ matrix $\mathbf{G}$

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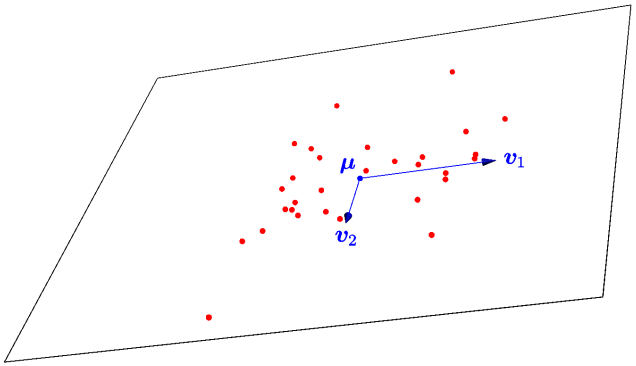
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Outline

1. Missing Data and Recommenders
2. Low-Rank Approximation
3. **Probabilistic Approach**
4. Non-Negative Matrix Factorisation



Bayes' Rule and MAP

- For parameters $\boldsymbol{\theta}$ and data $\mathcal{D}$, Bayes' rule gives the **posterior** (likelihood $\times$ prior / evidence)

$$\mathbb{P}(\boldsymbol{\theta}|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\boldsymbol{\theta})\mathbb{P}(\boldsymbol{\theta})}{\mathbb{P}(\mathcal{D})}$$

- The **maximum a posteriori** (MAP) solution is the most probable $\boldsymbol{\theta}$ ($\mathbb{P}(\mathcal{D})$ does not depend on it)

$$\boldsymbol{\theta}_{\mathrm{MAP}} = \operatorname*{argmax}_{\boldsymbol{\theta}} \Big(\log(\mathbb{P}(\mathcal{D}|\boldsymbol{\theta})) + \log(\mathbb{P}(\boldsymbol{\theta})) \Big)$$

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Bayesian Approach

- We can use a Bayesian approach to infer $\boldsymbol{A}$ and $\mathbf{B}$
- The likelihood is

$$\mathbb{P}(X_{ij}|\mathbf{A},\mathbf{B}) = \mathcal{N}\left(X_{ij}\middle|\sum_{k=1}^{K} A_{ik}B_{kj},\sigma^2\right)$$

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma}\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}$$

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Priors

- We also need priors

$$\mathbb{P}(\mathbf{A}) = \prod_{i,k} \mathcal{N}(A_{ik}|0,\sigma_A^2), \qquad \mathbb{P}(\mathbf{B}) = \prod_{k,j} \mathcal{N}(B_{kj}|0,\sigma_B^2)$$

- σ_A^2 and σ_B^2 explain the variation in the original matrix $\mathbf{X}$

- Note that

$$\mathbb{E}\left[\sum_{k=1}^{K} A_{ik}B_{kj}\right] = 0, \qquad \mathbb{V}\mathrm{ar}\left(\sum_{k=1}^{K} A_{ik}B_{kj}\right) = K\sigma_A^2\sigma_B^2$$

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MAP Solution

- Calculating the posterior probability for $\mathbf{A}$ and $\mathbf{B}$ is intractable
- We can, however, compute the *maximum a posteriori* solution

$$\mathcal{E} = \sum_{(i,j)\in\mathcal{D}} \log(\mathbb{P}(X_{ij}|\mathbf{A},\mathbf{B})) + \log(\mathbb{P}(\mathbf{A})) + \log(\mathbb{P}(\mathbf{B})) + C$$

$$= -\frac{1}{2\sigma^2}\sum_{(i,j)\in\mathcal{D}}\left(X_{ij} - \sum_{k=1}^{K} A_{ik}B_{kj}\right)^2 - \frac{1}{2\sigma_A^2}\sum_{i,k} A_{ik}^2$$

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Iterative Learning

- We can maximise the log-posterior by first maximising the log-posterior with respect to $\mathbf{A}$ keeping $\mathbf{B}$ fixed

$$\frac{\partial \mathcal{E}}{\partial A_{ik}} = \frac{1}{\sigma^2} \sum_{j|(i,j)\in\mathcal{D}} \left(X_{ij} - \sum_{l=1}^{K} A_{il} B_{lj} \right) B_{kj} - \frac{1}{\sigma_A^2} A_{ik} = 0$$

- This is equivalent to

$$\sum_{l=1}^{K} \left(\frac{\sigma^2}{\sigma_A^2} \delta_{k,l} + \sum_{j|(i,j)\in\mathcal{D}} B_{lj} B_{kj} \right) A_{il} = \sum_{j|(i,j)\in\mathcal{D}} B_{kj} X_{ij}$$

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Solving for $\boldsymbol{A}$

- We got

$$\sum_{l=1}^{K}\left(\frac{\sigma^2}{\sigma_A^2}\delta_{k,l}+\sum_{j|(i,j)\in\mathcal{D}}B_{lj}B_{kj}\right)A_{il}=\sum_{j|(i,j)\in\mathcal{D}}B_{kj}X_{ij}$$

- Which has the form $\mathbf{M}^i\boldsymbol{a}^i=\boldsymbol{c}^i$

$$M^i_{kl}=\frac{\sigma^2}{\sigma_A^2}\delta_{k,l}+\sum_{j|(i,j)\in\mathcal{D}}B_{lj}B_{kj},\qquad a^i_l=A_{il},\qquad c^i_k=\sum_{j|(i,j)\in\mathcal{D}}B_{kj}X_{ij}$$

- Solution is $\boldsymbol{a}^i=(\mathbf{M}^i)^{-1}\boldsymbol{c}^i$ (one $K\times K$ solve for each row i)

Solving for A

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Each Step is Ridge Regression

- Write $\boldsymbol{b}_j$ for column j of $\mathbf{B}$ and $\boldsymbol{a}_i$ for row i of $\boldsymbol{A}$ (as a column vector)
- With $\mathbf{B}$ fixed, row i of $\boldsymbol{A}$ minimises

$$\sum_{j|(i,j)\in\mathcal{D}} \left(X_{ij} - \boldsymbol{a}_i^\mathsf{T}\boldsymbol{b}_j\right)^2 + \eta\|\boldsymbol{a}_i\|^2, \qquad \eta = \frac{\sigma^2}{\sigma_A^2}$$

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Converging to a Solution

- Then solve for $\mathbf{B}$ (almost identical to solving for $\mathbf{A}$)
- Iterate between $\mathbf{A}$ and $\mathbf{B}$ until they converge
- No step can decrease the log-posterior, so it converges, usually rapidly (to a local optimum: the full problem is not convex)
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Embeddings

- Row i of $\mathbf{A}$ is a K-dimensional vector describing user i; column j of $\mathbf{B}$ describes item j
- The predicted rating is their inner product, $\boldsymbol{a}_i^\top \boldsymbol{b}_j$
- Items with similar vectors are rated alike: we have learnt an **embedding** of users and items
- Modern recommenders learn embeddings with neural networks, mostly from **implicit feedback** (clicks, views, purchases): a missing entry is then weak evidence of not liking, not simply unknown
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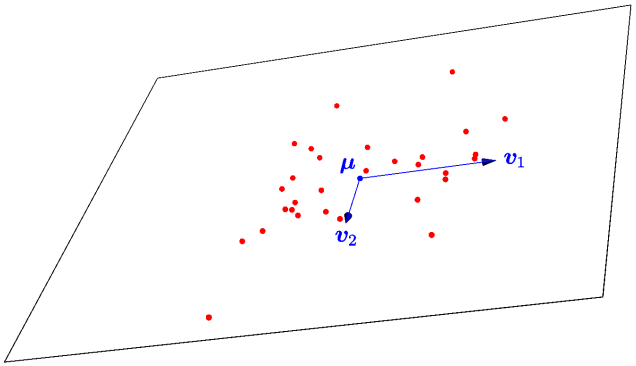
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Outline

1. Missing Data and Recommenders
2. Low-Rank Approximation
3. Probabilistic Approach
4. **Non-Negative Matrix Factorisation**



Matrix Factorisation

- Although we have treated matrix factorisation as a matrix completion task we can also think of it as a data modelling task
- Consider a set of books in mathematics, physics and biology
- We can build a matrix of books (columns) versus words (rows), with elements being the word count
- Can use the SVD, taking $\mathbf{A}$ as the first three columns of $\mathbf{US}$ and $\mathbf{B}$ as the first three rows of $\mathbf{V}^{\mathsf{T}}$

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SVD

	Euclid	Gauss	Feynman	Newton	Darwin	Mendel
theorem	51	101	5	10	3	1
proof	51	103	2	8	0	0
sum	22	51	60	72	5	7
vector	0	70	90	50	3	3
wave	0	1	51	92	2	0
species	0	4	5	1	161	89
fitness	0	1	3	1	35	2

$\approx$

	1	2	3
theorem	9.0	-2.2	-6.6
proof	8.8	-2.5	-7.0
sum	10.2	-1.7	3.4
vector	11.2	-2.3	3.4
wave	6.5	-1.3	7.7
species	5.2	17.7	-0.5
fitness	1.0	3.0	0.0

	Euclid	Gauss	Feynman	Newton	Darwin	Mendel
1	2.5	6.9	4.5	4.5	2.2	1.3
2	-0.8	-1.9	-0.9	-1.1	8.5	4.5
3	-3.5	-5.6	4.9	5.7	-0.3	-0.1

- Best rank-3 approximation, but the factors mix signs: hard to interpret

Non-Negative Matrix Factorisation

- When $\mathbf{X}$ is non-negative we can get a sparse model by minimising $E(\boldsymbol{A},\mathbf{B}) = \|\mathbf{X} - \boldsymbol{A}\mathbf{B}\|_F^2$ with all elements of $\boldsymbol{A}$ and $\mathbf{B}$ non-negative
- Projected gradient descent: take a gradient step, then set negative elements to zero
- Or Lee and Seung's **multiplicative updates** ($\oslash$ is element-by-element division)

$$\mathbf{B} \leftarrow \mathbf{B} \odot \left(\boldsymbol{A}^\mathsf{T}\mathbf{X}\right) \oslash \left(\boldsymbol{A}^\mathsf{T}\boldsymbol{A}\mathbf{B}\right), \quad \boldsymbol{A} \leftarrow \boldsymbol{A} \odot \left(\mathbf{X}\mathbf{B}^\mathsf{T}\right) \oslash \left(\boldsymbol{A}\mathbf{B}\mathbf{B}^\mathsf{T}\right)$$

which never make an element negative

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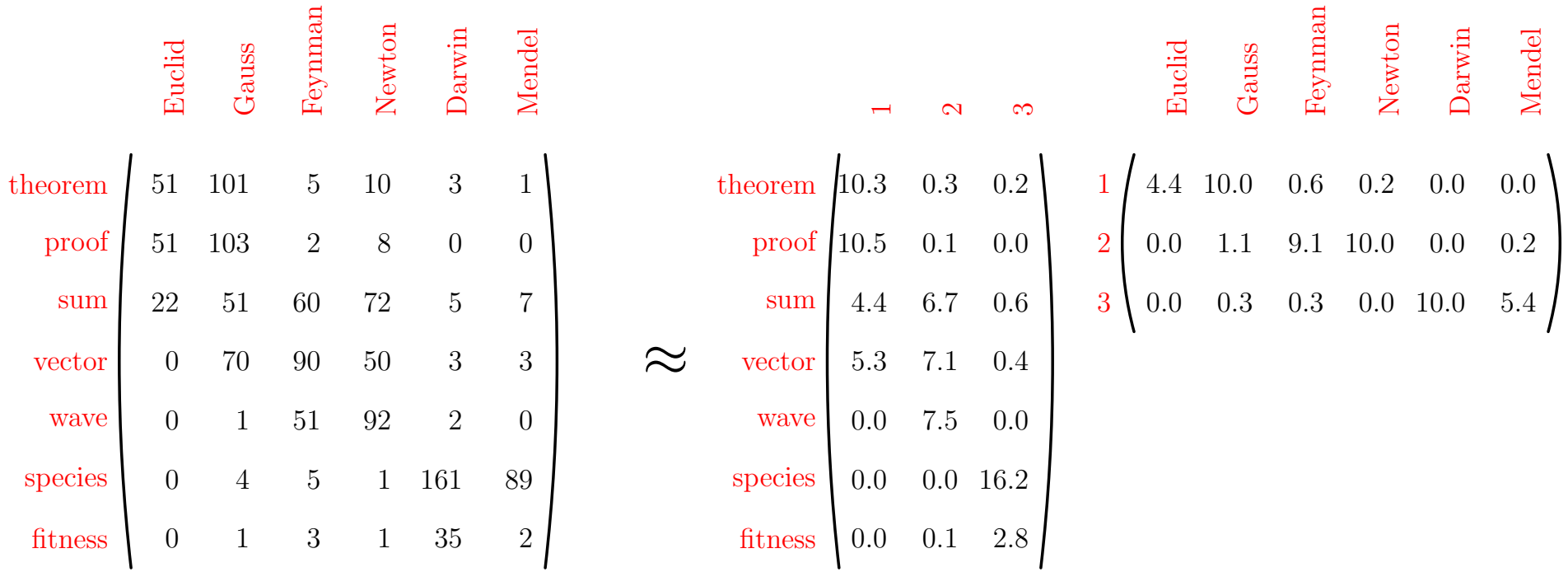
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species	0.0	0.0	16.2
fitness	0.0	0.1	2.8

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2	0.0	1.1	9.1	10.0	0.0	0.2
3	0.0	0.3	0.3	0.0	10.0	5.4

Non-Negative Matrix Factorisation



- The three factors pick out mathematics, physics and biology

Non-Negative Matrix Factorisation: Caveats

- NMF is very popular because its factors are often easier to interpret than those of the SVD
- But
  - ⋆ It cannot fit better than the SVD: by Eckart–Young the truncated SVD is the best rank-K fit (here relative errors 0.164 against 0.158)
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Conclusion

- Structured data typically lies close to a low-dimensional subspace, and we can exploit this
- Matrix factorisation has many applications including recommender systems, data completion, and data modelling
- The truncated SVD gives the best low-rank approximation; iterating it fills in missing entries
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